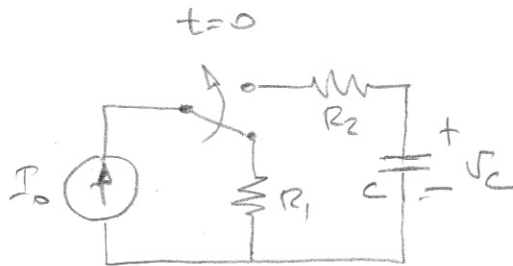


5.32)



$$I_0 = 17 \text{ mA}$$

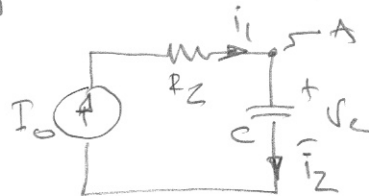
$$C = 0.55 \mu\text{F}$$

$$R_1 = 7 \text{ k}\Omega$$

$$R_2 = 3.3 \text{ k}\Omega$$

$$V_C(t=0) = -7 \text{ V. BESTÄTIG V}_C(t > 0).$$

Lösung: für $t \geq 0$ Kuer



$$\text{für nod A Kuer KSTL, } i_1 - i_2 = 0 \quad (1)$$

$$\text{MEN } i_1 = I_0 \text{ oder } i_2 = C \frac{dV_C}{dt} \quad (2)$$

(2) i(1) Kuer

$$I_0 - C \frac{dV_C}{dt} = 0$$

Integriert MAP t

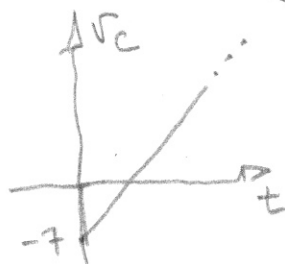
$$\int_{V_C(0)}^{V_C(t)} dV_C = \int_0^t \frac{I_0}{C} dt$$

$$\Rightarrow V_C(t) - V_C(0) = \frac{I_0}{C} t$$

$$V_C(t) = V_C(0) + \frac{I_0}{C} t$$

MEDSIFPROZ

$$V_C(t) = -7 + 30.91 \cdot 10^3 t \text{ VOLT}$$



SPÄNNUNGEN SIEHER MONOTONT
FÜR ALLE $t > 0$. EJ REALISTISCHT.

$$\underline{\text{SVAR:}} V_C(t) = -7 + 30.91 \cdot 10^3 t \text{ V}$$